AUGUST MATH PREDICTION

32 Hard Math Questions

Digital DSAT Math — Hard Level Practice

DSAT ADVANTAGE

Full question set with realistic math typesetting and figures

Question 1

A quadratic function models the height of an object above the ground. The object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. What was the height of the object 13 seconds after it was launched?

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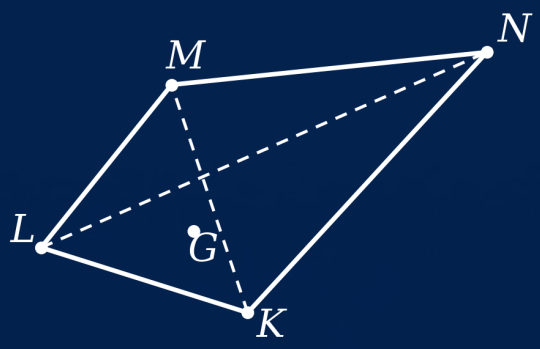
Question 2

$$y = 900ax^2$$

For a certain type of rope, the equation above gives the estimated breaking strength y (in pounds) of a rope with circumference x (in inches). If a rope with circumference of 2.75 inches has breaking strength of 9,528.75 pounds, what is the breaking strength of a rope with circumference of 8.50 inches?

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Question 3



In quadrilateral KLMN, KL = 3, LM = 3, KN = 27, and MN = 27. Diagonals intersect at point G, where GK = 1 and GM = 1. If the length of diagonal LN is $\sqrt{p}+\sqrt{w}$, what is p + w? (Figure not drawn to scale.)

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Question 4

$$c(x-6)=-7(x+k)$$

In the given equation, c and k are constants. The equation has exactly one solution. Which statement must be true?

A. The value of c cannot be –7.

B. The value of c cannot be –7/6.

C. The value of k cannot be 6/7.

D. The value of k cannot be –6.

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Question 5

During the first 8.00 seconds, a car's speed increased to 11.0 m/s. From 8.00 to 14.0 seconds, speed increased from 11.0 m/s to 23.0 m/s. What is the positive difference between the average rates of change of speed during these two intervals? Round your answer to the nearest hundredth.

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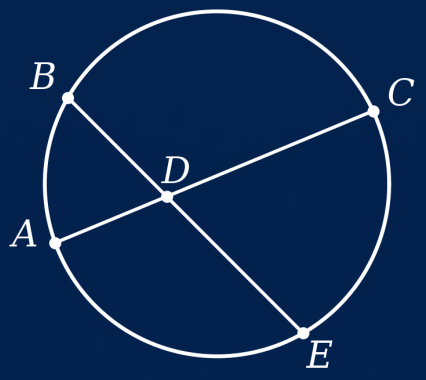
Question 6

$$(x-k)^2=(k-4a)(x-k)$$

In the given equation, a and k are constants with k > 4a. The sum of solutions is 3k + 37. What is the value of a?

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Question 7



In the figure shown, points A, B, C, and E lie on the circle, and AB < BC. Segment AC is perpendicular to segment BE at point D, and BD = $\sqrt{322}$. The diameter of the circle is 163. If CD/AD = r, what is the value of r?

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Question 8

Which expression is NOT a factor of $7{,}290x^4 - 56{,}250$?

A. $9x^2 + 25$

B. $3x^2 - 5$

C. $3x + 5$

D. 90

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Question 9

Which of the following expressions has 6x – 5b as a factor, where b is a positive integer constant?

A. $24x^2 – 22x – 15b$

B. $24x^2 + 18x – 15b$

C. $24x^2 + 36x – 15b$

D. $24x^2 – 484x – 15b$

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Question 10

A rectangular area consists of 1372 equal squares where each square has an area of k. If the length of the rectangular area is 1.75 times the width and the length is equal to $x\sqrt{k}$, what is the value of x?

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Question 11

The expression $6x^4 + 23x^2 + 7$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of a + c?

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Question 12

$$f(x) = -x^3 - ax^2 - bx - c$$

The function f is defined by the equation above, where a, b, and c are integer constants. The zeros of the function are –9, –4, and 5. What is the value of a + b?

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Question 13

For two acute angles, ∠Q and ∠R, cos Q = sin R. The measures, in degrees, of ∠Q and ∠R are ∠Q = 4x + 9° and ∠R = 4x – 3°, respectively. What is the value of x?

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Question 14

$$\frac{5}{16}x + \frac{7}{16}y = \frac{8}{7} - \frac{7}{14}y$$

$$\frac{5}{6}x + \frac{7}{6} = py + \frac{3}{4}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

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Question 15

$$h(z) = z^4 + az^3 + bz^2 + cx + d$$

The function h(z) is defined by the equation above, where a, b, c, and d are constants. In the xy-plane, the graph of y = h(x) passes through the points (–6, 0), (–2, 0), (4, 0), and (5, 0). If h(0) = q and the product of all the values of x for which h(x) = 0 is r, where q and r are constants, what is the value of q + r?

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Question 16

An object's speed, in meters per hour, is increasing at a rate of 1,490 meters per hour per second. Which of the following is closest to this rate in feet per second squared? (Use 1 meter = 3.28 feet)

A) 0.126

B) 1.36

C) 7.57

D) 81.5

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Question 17

$$x = \sqrt[2n]{7x^n + 30}$$

Solve for x in the given equation. Find the value of x^n.

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Question 18

The function f(x) = (x – a)(x – b) is defined by the given equation, where a and b are integer constants. If f(32) > 0, f(35) < 0, and f(38) > 0, which of the following could be the value of a + b?

A) 35

B) 37

C) 67

D) 69

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Question 19

$$f(s) = a \cdot b^s$$

The exponential function f is defined by the equation above, where a and b are positive constants. If f(s) = t and f(s + 1) = t – 0.87t, where s and t are constants, what is the value of b?

A) 0.13

B) 0.87

C) 1.13

D) 1.87

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Question 20

$$\frac{x^{27}(x-7)}{3x^3}+\frac{7x^{27}}{3x^3}=\frac{1}{3}x^k$$

The expression above is equivalent to $(1/3)x^k$, where k is a constant and $x > 0$. What is the value of k?

A) 3

B) $\frac{27}{3}$

C) 25

D) 28

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Question 21

(x + 7) is a factor of the quadratic function y = f(x), and x = c is a positive zero of the function. If the minimum of the function is (63/8, k), what is the value of c?

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Question 22

$$g(x) = ab^{\frac{x}{n}}$$

In the given function, g(6) = 11 and g(9) = 704. What is the value of g(11)?

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Question 23

$$f(x) = a^x - b$$

The exponential function given above passes through points (c, 8) and (2c, 350). If c is a positive constant, what is a possible value of b?

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Question 24

The speed of object K is 9,525% of the speed of object J, and the speed of object K is 0.025% of the speed of object L. If the speed of object L is x% of the speed of object J, what is the value of $\frac{x}{1{,}000}$?

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Question 25

$$f(x) = ax^2 + bx + c$$

The function f is defined by the equation above, where a, b and c are constants. The graph of y = f(x) in the xy-plane passes through the points (18, 0) and (–11, 0). If a is an integer greater than 1, which of the following could be the value of a + b?

A) –12

B) –6

C) 7

D) 14

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Question 26

$$j(x) = \frac{10x + 5}{3p}$$

$$k(x) = \frac{j(x) \cdot p}{x + 1}$$

The function j is defined above, and the function k is defined above, where p is a positive constant. What is the value of k(9)?

A. $\frac{19}{6}$

B. $\frac{10}{3}$

C. $\frac{95}{13}$

D. $\frac{85}{3}$

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Question 27

$$\frac{|4x - 48| + 4}{8} = 6$$

What is the sum of the solutions to the given equation?

A. 1

B. 23

C. 24

D. 48

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Question 28

$$y = 3\left(\frac{a}{6}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy-plane cross the x-axis, where a, b, and c are positive constants such that a > 6 and b > c?

A. Zero

B. One

C. Two

D. Three

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Question 29

$$2x^2 - 8x - 21 = 0$$

One solution to the given equation can be written as $x = h - \frac{1}{2}\sqrt{k}$, where h and k are integer constants. What is the value of h + k?

A. -24

B. 10

C. 29

D. 60

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Question 30

$$v(x) = \frac{x^2 + bx + c}{(x + 5)(x - 19)}$$

The function v is defined by the equation above, where b and c are constants. In the xy-plane, the graph of y = v(x) passes through the points (0, 66/95) and (11, 0). If v(q) = 0, which of the following could be the value of q?

A. -6

B. -5

C. 5

D. 6

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Question 31

$$f(x) = 55(0.19)^x$$

The function f is defined by the equation above. For any positive integer n, the value of f(n) is p% less than the value of f(n – 1). What is the value of p?

A. 19

B. 45

C. 55

D. 81

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Question 32

$$x^2 + y^2 = 36 \qquad y = mx + \frac{b}{4}$$

In the given system of equations above, m and b are negative constants. In the xy-plane, the graphs of the equations in the given system intersect at the point (–5, y), where y < 0. Which expression represents the value of b?

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

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